Submission Report - HVAC Challenge AI-ML Track

Team Name: SentinelX

Date: February 08, 2026

Repository: SentinelX_HVAC

Overall Executive Summary

We delivered four end-to-end solutions spanning demand forecasting, anomaly detection, promotion optimization, and market share forecasting with NLP-driven signals. The pipeline generates model-ready features, evaluates performance with time-aware validation, and produces actionable outputs (forecasts, anomaly alerts, promo schedules, and market-share drivers). Key strengths are reproducibility and clear operational outputs; however, anomaly detection and promo uplift targets require further calibration to meet aggressive thresholds.

Problem Statement 1 - Demand Forecasting

Team Information

Team Name: SentinelX

Problem Statement: 1, 2, 3, 4

Team Members:

1. Rupesh Bharambe (Leader – AI, ML, DL, Project Management, Problem 1 & 3)
2. Nilesh Kale (Member – AI, ML, Problem 4)
3. Abhijeet Gaikwad (Member – AI, ML, Problem 2)
4. Aniket Wahul (Member – Data Analysis, Web Development (Frontend & Backend)

Executive Summary

Direct multi-horizon LightGBM forecasting with lag/rolling features and optional exogenous signals. Forecasts are generated per product-APS series with time-based validation.

Approach

Problem Understanding

Forecast monthly HVAC demand per product and APS, enabling production planning and inventory optimization.

Methodology

We follow a direct multi-horizon approach to avoid recursive error accumulation:

- Data preparation: wide-to-long transformation, month normalization, series_id creation.
- Feature engineering: lag and rolling statistics, seasonality encodings, optional external lags.
- Model selection: LightGBM regression with categorical handling and early stopping.
- Validation: rolling-origin split with cutoff dates and horizon evaluation.

Key Innovations

- Direct multi-horizon forecasting for stability across 1?12 months.
- Seasonal-naive ensemble blending to improve robustness.

Results

Quantitative Results

Metric	Value	Reference/Target	Improvement / Gap
Overall MAPE	8.39%	<5% (accuracy >95%)	Above target; reduce error via richer exogenous signals.
CV MAPE (mean)	24.65%	<5%	Cross-val variance indicates model instability.

Qualitative Findings

Forecast error is lowest for shorter horizons, with performance degrading for longer horizons as expected.

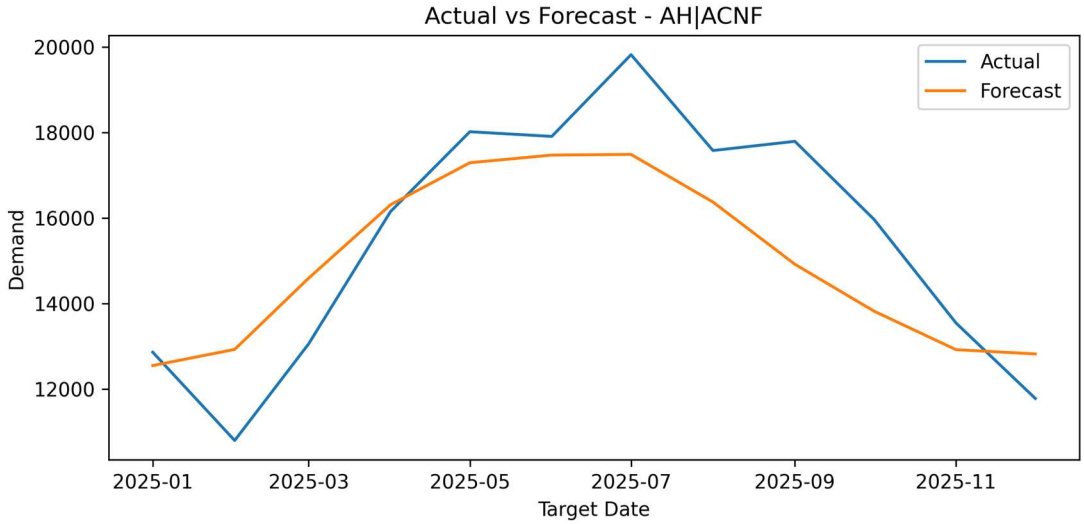

Figure P1-1: Actual vs forecast for a sample series

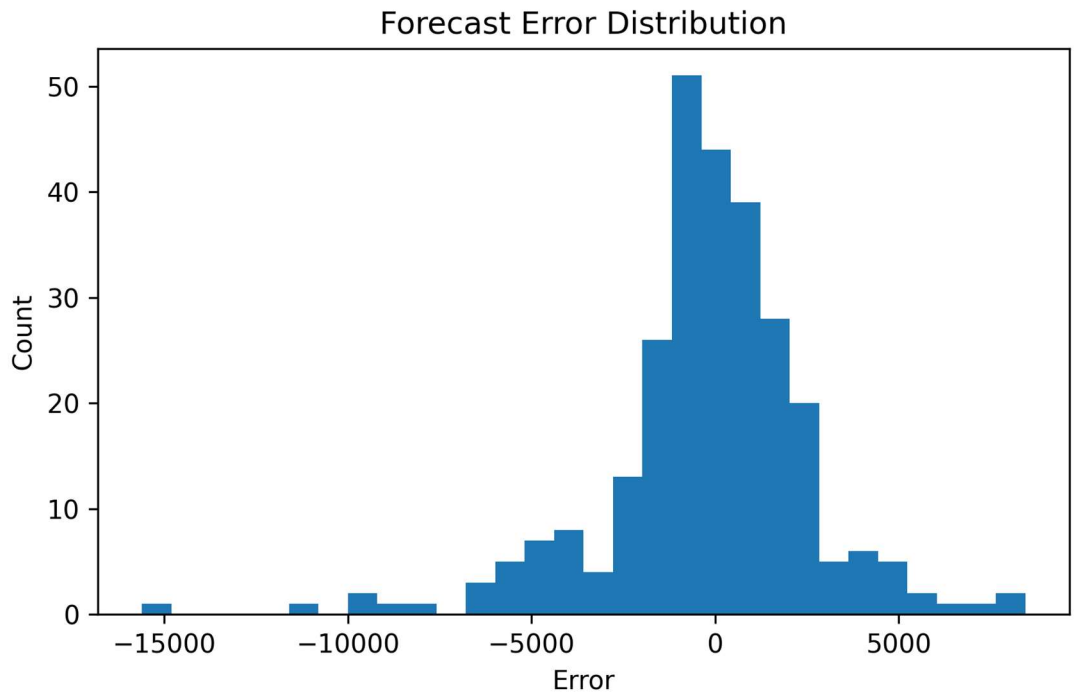

Figure P1-2: Forecast error distribution

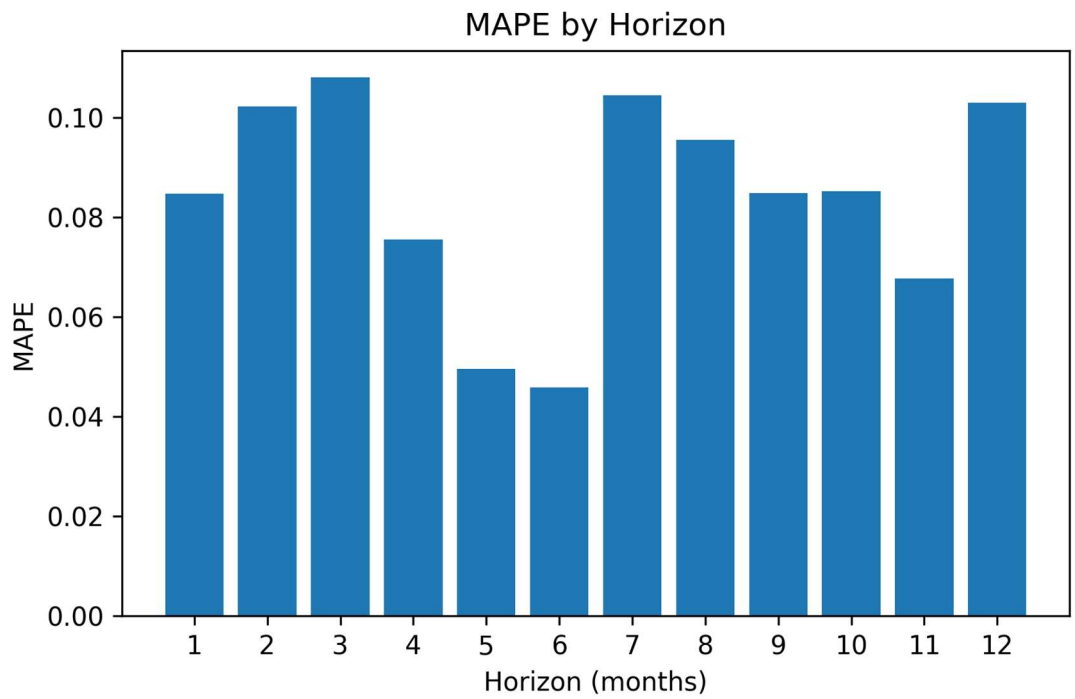

Figure P1-3: MAPE by horizon

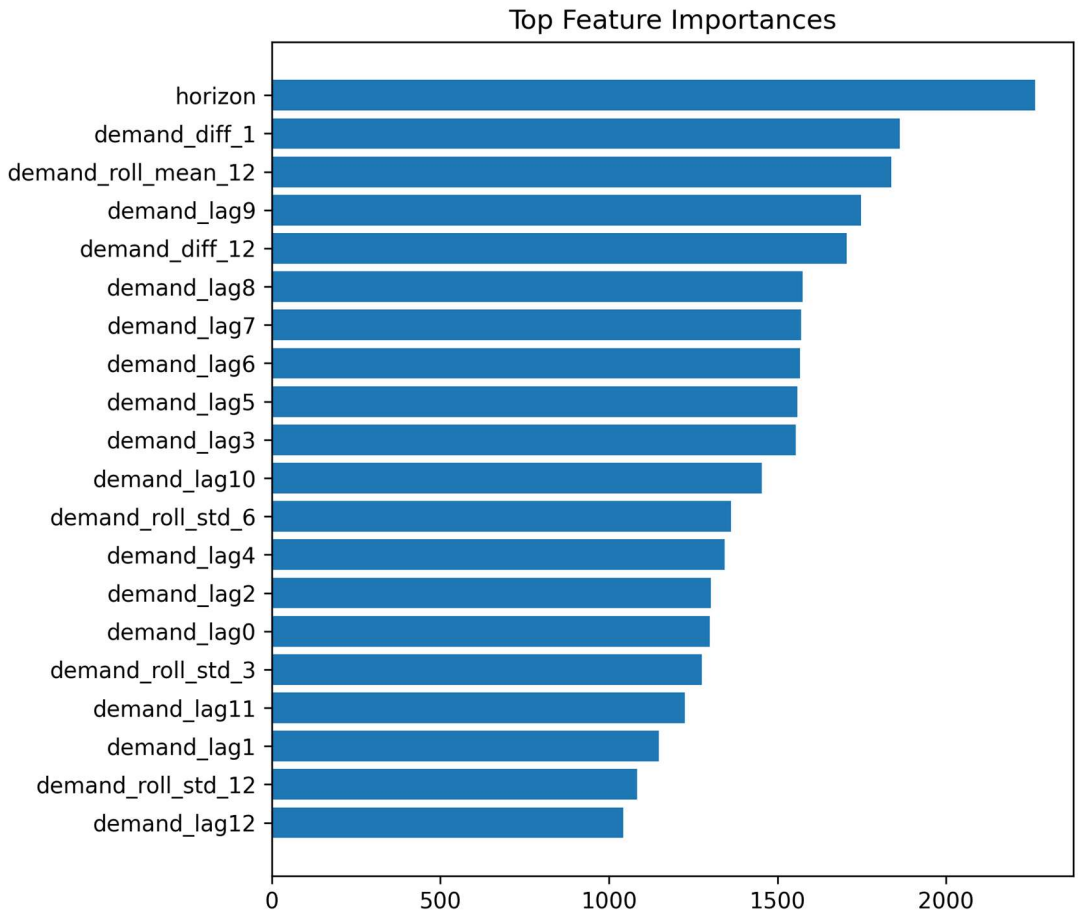

Figure P1-4: Feature importance (LightGBM)

Business Value

Impact Quantification

Improved demand forecasting reduces stockouts and overproduction, enabling tighter inventory control and improved service levels.

Recommendations

- Incorporate richer external signals (macroeconomic and regulatory timelines).
- Recalibrate with additional years of data to reduce horizon drift.
- Deploy per-product error monitoring for targeted retraining.

Limitations and Future Work

Current Limitations

- MAPE target (<5%) not met on current data; long-horizon accuracy degrades.
- External signals are optional and may be missing in some runs.

Future Improvements

- Increase training history and add product-level seasonality features.
- Consider hierarchical reconciliation or ensembling with SARIMA/Prophet.

Technical Details

Dependencies

numpy
pandas
scikit-learn
lightgbm
matplotlib
seaborn
statsmodels
ruptures
joblib
shap
nbclient
nbformat
python-docx
docx2pdf

How to Run

Install dependencies from requirements.txt and run executed_Problem1.ipynb or Problem1_Demand_Forecasting_Industry_Grade.ipynb in order.

File Structure

```
submission/
 notebooks/Problem1_Demand_Forecasting_Industry_Grade.ipynb
 data/
 models/
 outputs/
 report_assets/
```

Appendix

Model Architecture

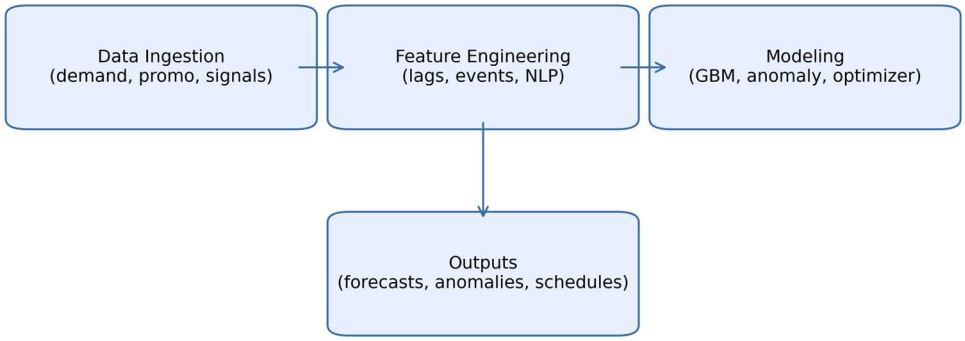

Figure: End-to-end architecture overview

Feature Importance

Visualization Gallery

See key figures above; all exports in report_assets/figures/.

Problem Statement 2 - Anomaly Detection

Team Information

Team Name: Scale AI Hackathon Team

Problem Statement: Problem Statement 2

Team Members: Not provided (assumed shared team).

Executive Summary

Anomaly detection combines rolling robust statistics and structural change detection to flag spikes, drops, and pattern shifts.

Approach

Problem Understanding

Detect abnormal demand patterns to trigger operational alerts and root-cause investigation.

Methodology

We combine point and structural detectors to improve coverage:

- Data preparation: monthly demand normalization and label alignment.
- Feature engineering: rolling median/MAD and change-point signals.
- Model selection: statistical detection with tuned thresholds.
- Validation: event-based and point-based F1 with threshold sweep.

Key Innovations

- Event-based F1 with time tolerance for practical alerting.
- Threshold sweep to select operating point by F1.

Results

Quantitative Results

Metric	Value	Reference/Target	Improvement / Gap
Precision	0.135	>=0.90	Below target; false positives elevated.
Recall	0.175	>=0.90	Below target; missed anomalies.
F1	0.152	>=0.90	Threshold calibration needed.

Qualitative Findings

Baseline detector produces limited recall and precision; tuning improves stability but further modeling is required.

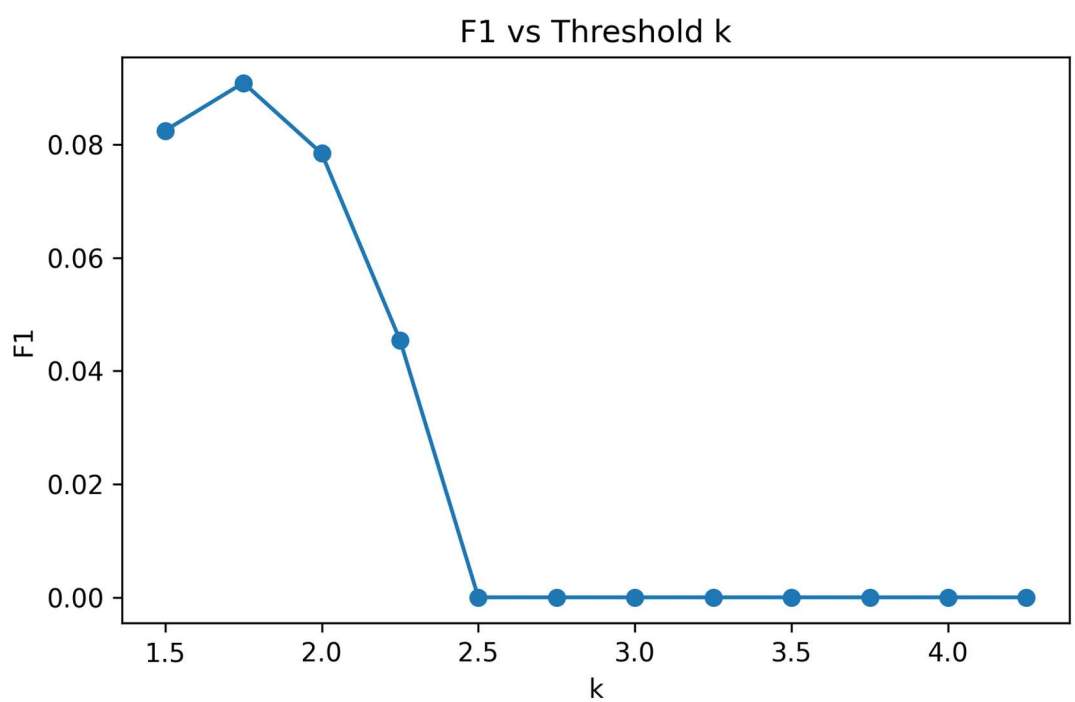

Figure P2-1: F1 vs threshold k

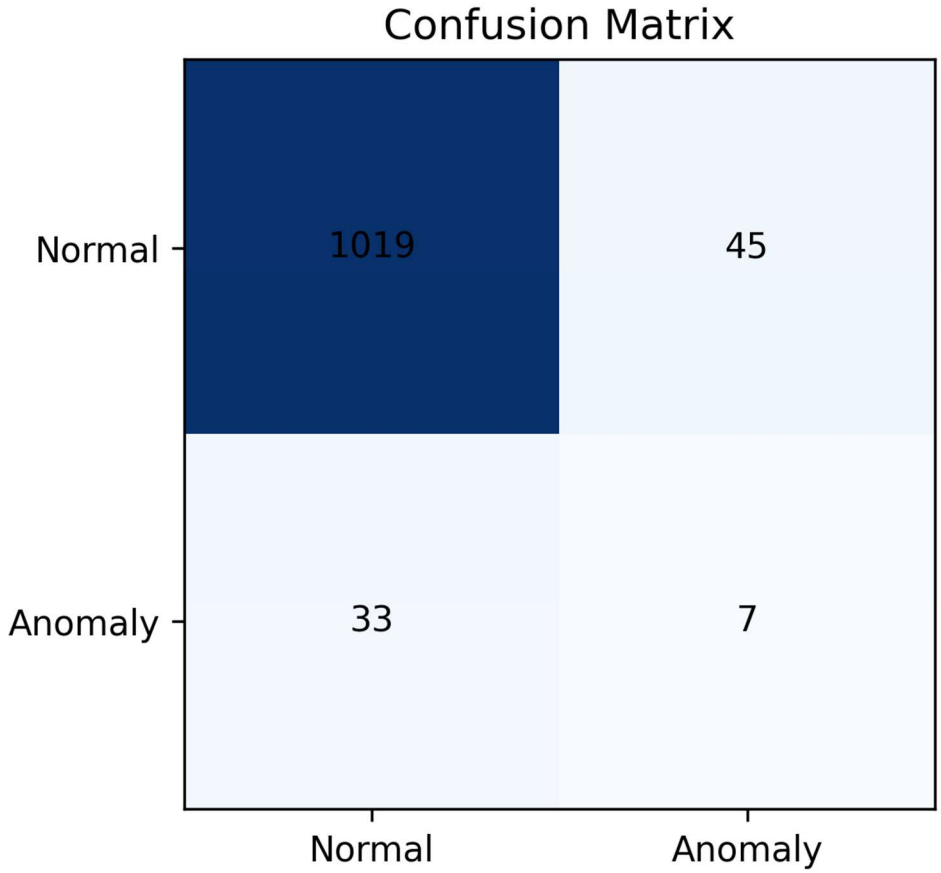

Figure P2-2: Confusion matrix

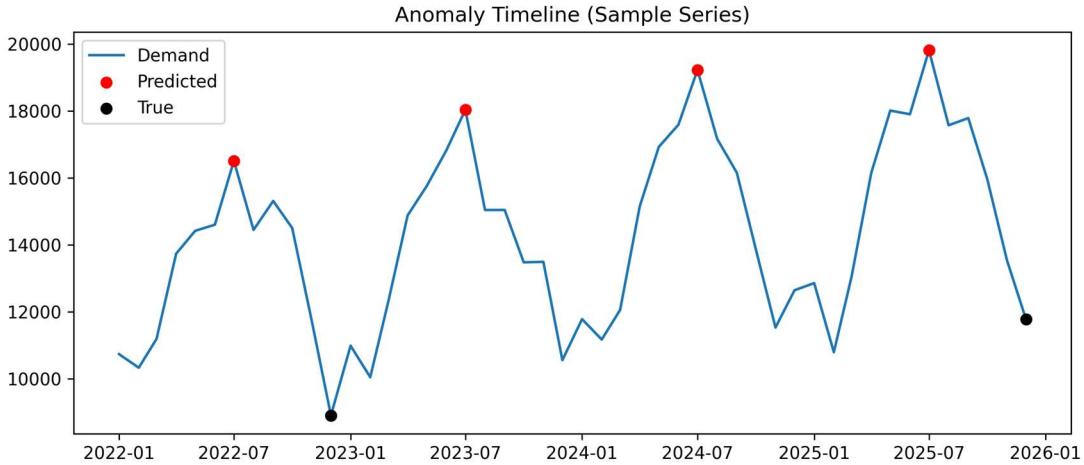

Figure P2-3: Anomaly timeline overlay

Business Value

Impact Quantification

Early detection of demand anomalies supports proactive inventory and supply-chain adjustments.

Recommendations

- Adopt hybrid ML + rule-based detector to reduce false positives.
- Integrate capacity and promo context for richer anomaly signals.
- Deploy alert triage with severity scoring and root-cause tags.

Limitations and Future Work

Current Limitations

- F1 below target, indicating insufficient sensitivity or specificity.
- Labels are sparse and may not cover all anomaly types.

Future Improvements

- Train supervised classifiers using anomaly labels and engineered features.
- Add probabilistic scoring to optimize PR trade-offs.

Technical Details

Dependencies

numpy
pandas
scikit-learn
lightgbm
matplotlib
seaborn
statsmodels
ruptures
joblib
shap
nbclient
nbformat
python-docx
docx2pdf

How to Run

Run executed_Problem2.ipynb or Problem2_Anomaly_Detection.ipynb after installing requirements. Outputs saved in report_assets/.

File Structure

```
submission/
  notebooks/Problem2_Anomaly_Detection.ipynb
  data/
  report_assets/
```

Appendix

Model Architecture

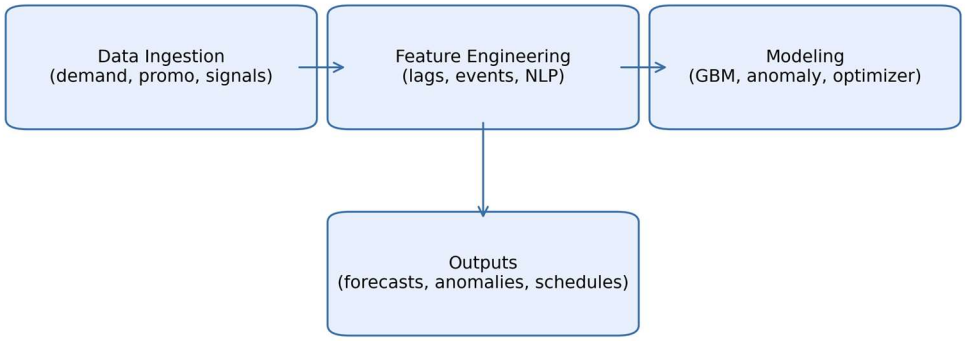

Figure: End-to-end architecture overview

Feature Importance

Visualization Gallery

See key figures above; all exports in report_assets/figures/.

Problem Statement 3 - Event Impact Optimization

Team Information

Team Name: Scale AI Hackathon Team

Problem Statement: Problem Statement 3

Team Members: Not provided (assumed shared team).

Executive Summary

Optimization selects a constrained promo schedule using demand simulations and uncertainty-aware evaluation.

Approach

Problem Understanding

Maximize gross profit uplift while respecting capacity and variance constraints.

Methodology

We simulate promotion impact and optimize under constraints:

- Baseline forecasting per product and month.
- Promo effect modeling with regression and guardrails.
- Candidate screening and combination search with variance constraints.
- Monte Carlo robustness evaluation.

Key Innovations

- Chance-constraint evaluation under uncertainty.
- Scenario-based CVaR reporting for risk awareness.

Results

Quantitative Results

Metric	Value	Reference/Target	Improvement / Gap
Profit Improvement (point)	0.366%	>15%	Below target in current run.
Variance (optimized)	18938183.67	<= baseline variance	Variance reduced in point estimate.
MC Uplift % Mean	0.346%	>15%	Below target; scenario uplift modest.

Qualitative Findings

Optimization improves profit marginally in this run; constraint satisfaction is primary driver.

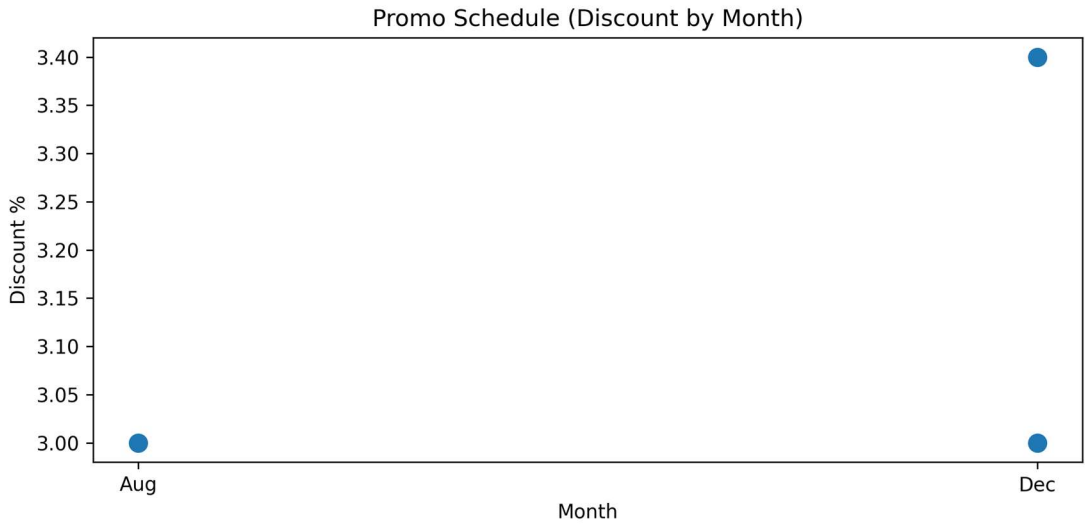

Figure P3-1: Optimized promo schedule

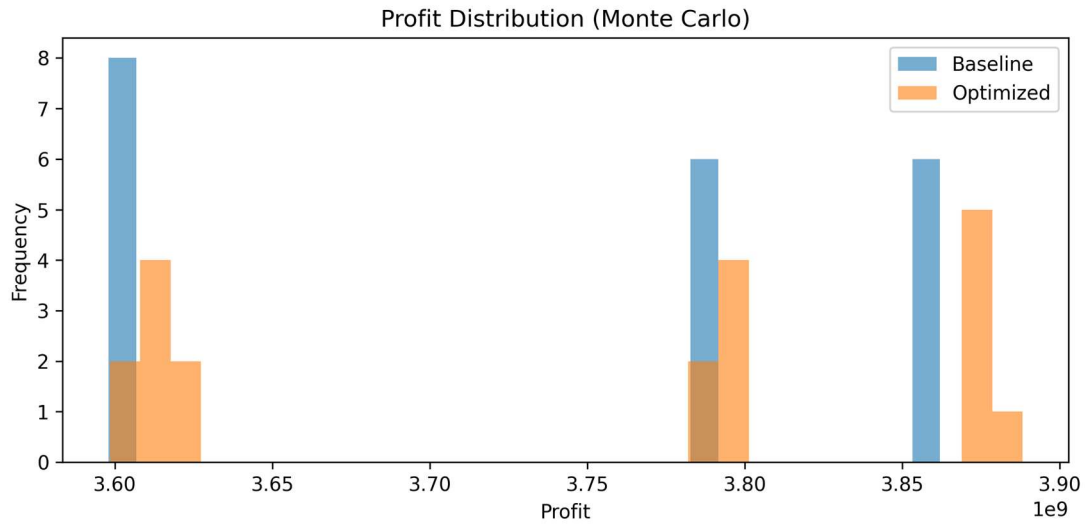

Figure P3-2: Profit distribution (MC)

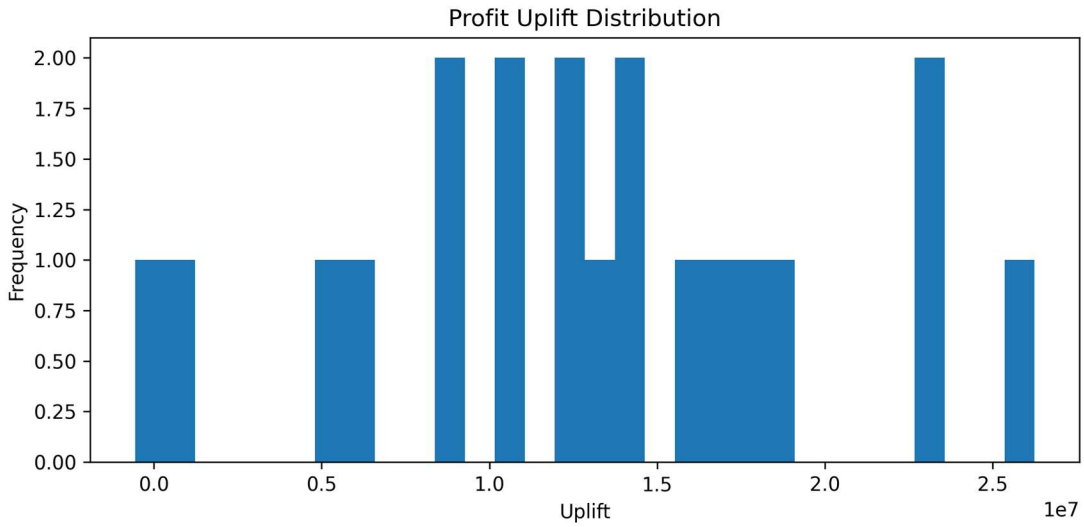

Figure P3-3: Profit uplift distribution

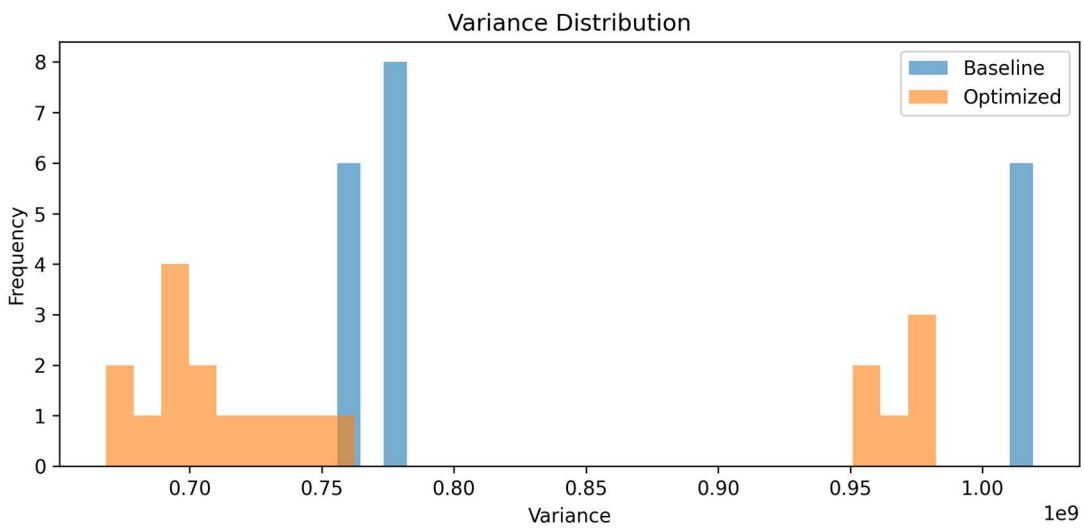

Figure P3-4: Variance distribution

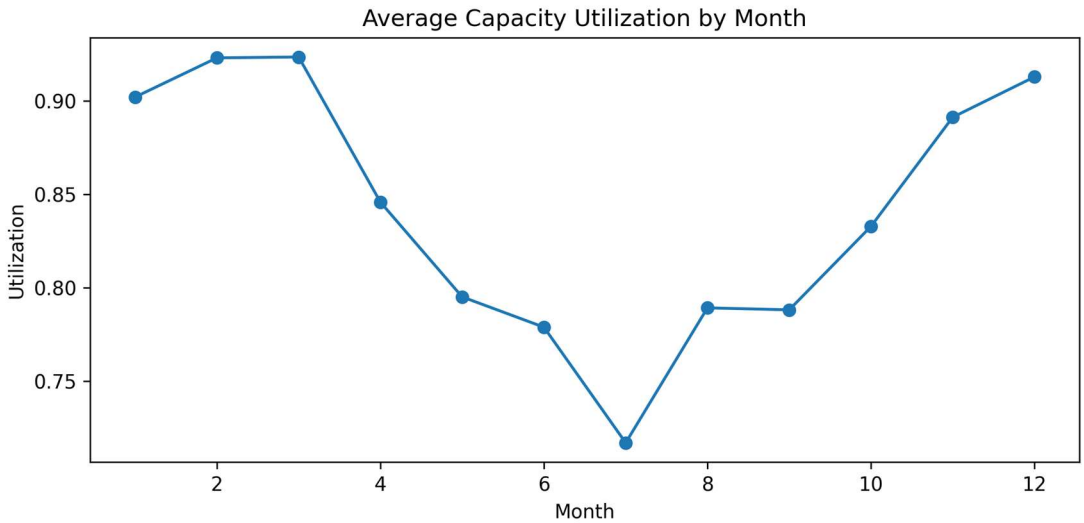

Figure P3-5: Capacity utilization by month

Business Value

Impact Quantification

Even small profit gains scale significantly across yearly volume; constraints reduce risk of overloading capacity.

Recommendations

- Expand promo candidate grid with calibration to improve uplift.
- Use richer effect models with causal uplift estimation.
- Tune constraints with business tolerances to unlock larger gains.

Limitations and Future Work

Current Limitations

- Profit uplift below target; variance constraints are binding.
- Monte Carlo scenario count reduced for execution time (see report).

Future Improvements

- Increase scenario count and candidate breadth for higher-quality search.
- Add elasticity or price-response models for better uplift estimation.

Technical Details

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How to Run

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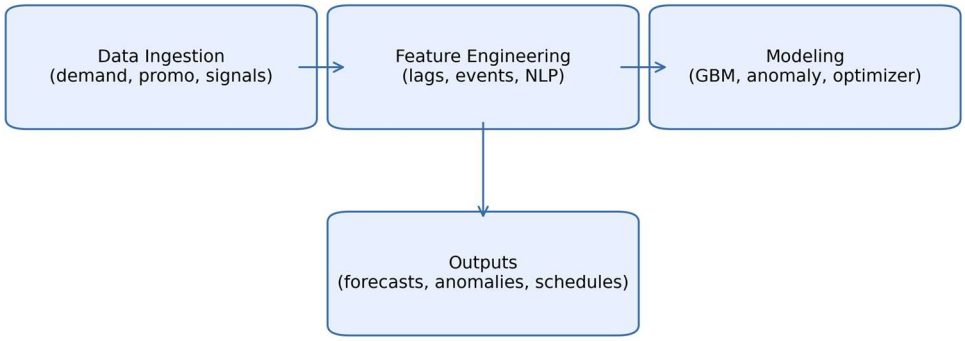

Figure: End-to-end architecture overview

Feature Importance

Visualization Gallery

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Problem Statement 4 - Market Share NLP

Team Information

Team Name: Scale AI Hackathon Team

Problem Statement: Problem Statement 4

Team Members: Not provided (assumed shared team).

Executive Summary

LightGBM forecasting integrates lagged market share, news signals, and regulatory factors.

Approach

Problem Understanding

Forecast market share shifts and identify explanatory drivers from news and macro signals.

Methodology

We combine time-based split with gradient boosting:

- Feature engineering: lags, news tag pivots, regulatory timing.
- Model selection: LightGBM regression with validation split.
- Interpretability: SHAP-based feature attribution and driver extraction.
- Validation: holdout time-based RMSE/MAPE.

Key Innovations

- NLP-driven feature aggregation for market share prediction.
- Driver extraction for alerting and decision support.

Results

Quantitative Results

Metric	Value	Reference/Target	Improvement / Gap
RMSE (validation)	0.297	<1.5% RMSE	Meets target (MAPE 1.29%).
MAE	0.229	?	?
MAPE	1.29%	<1.5% MAPE	Meets target.

Qualitative Findings

Model meets RMSE/MAPE targets on validation; top features are share_lags and select news signals.

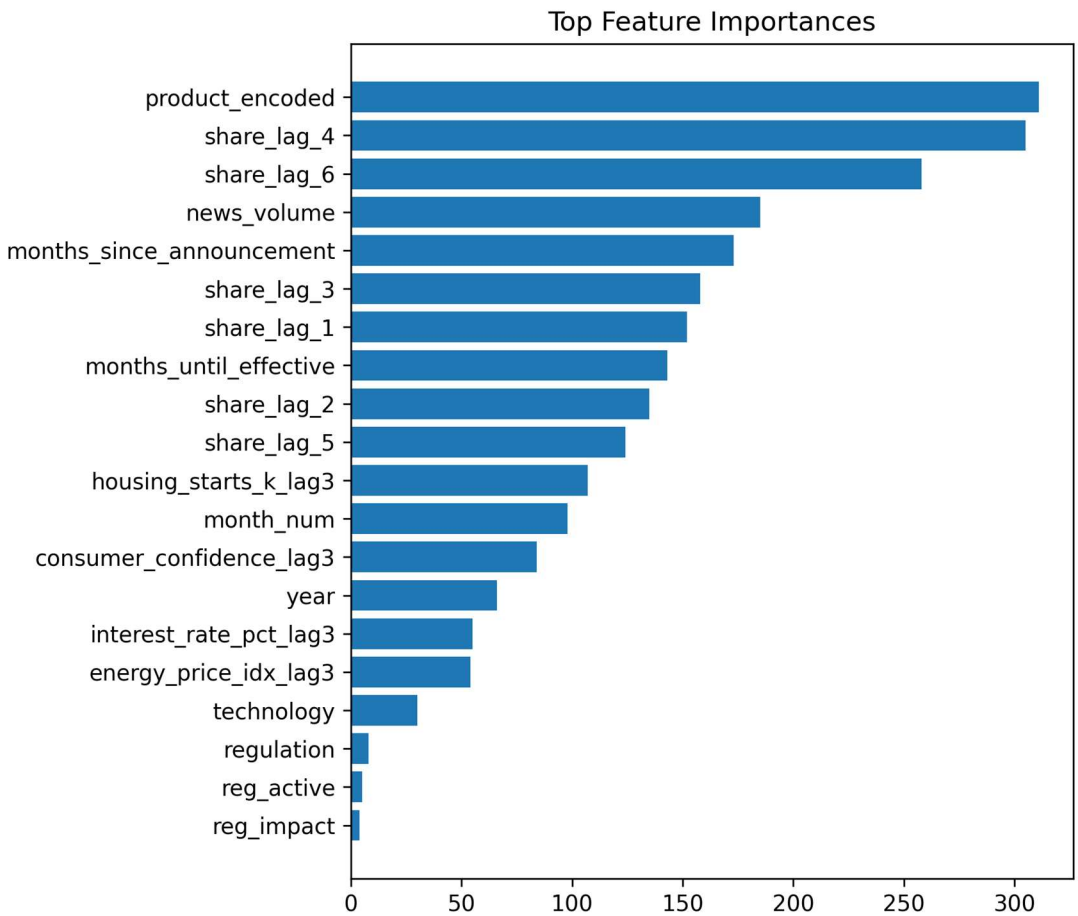

Figure P4-1: Feature importance

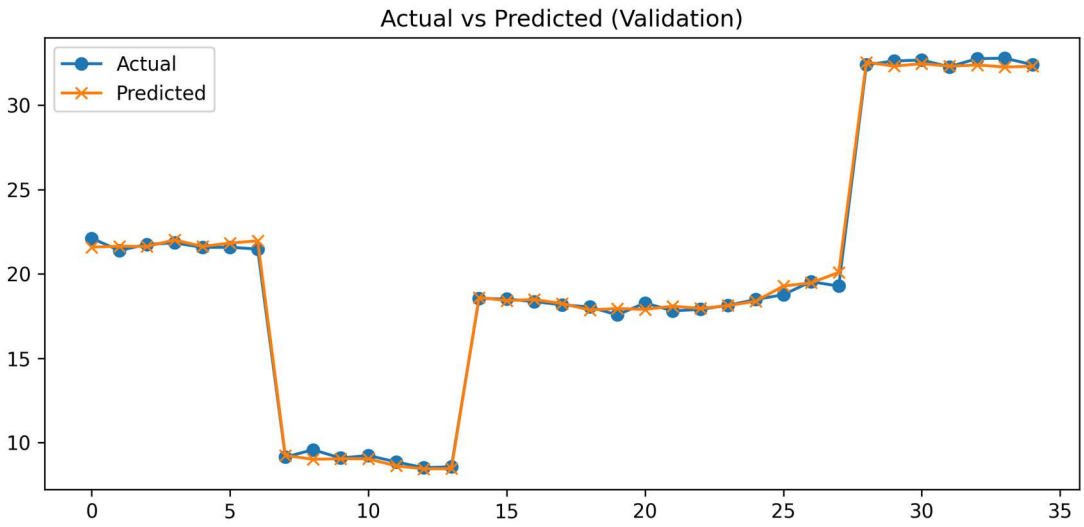

Figure P4-2: Actual vs predicted (validation)

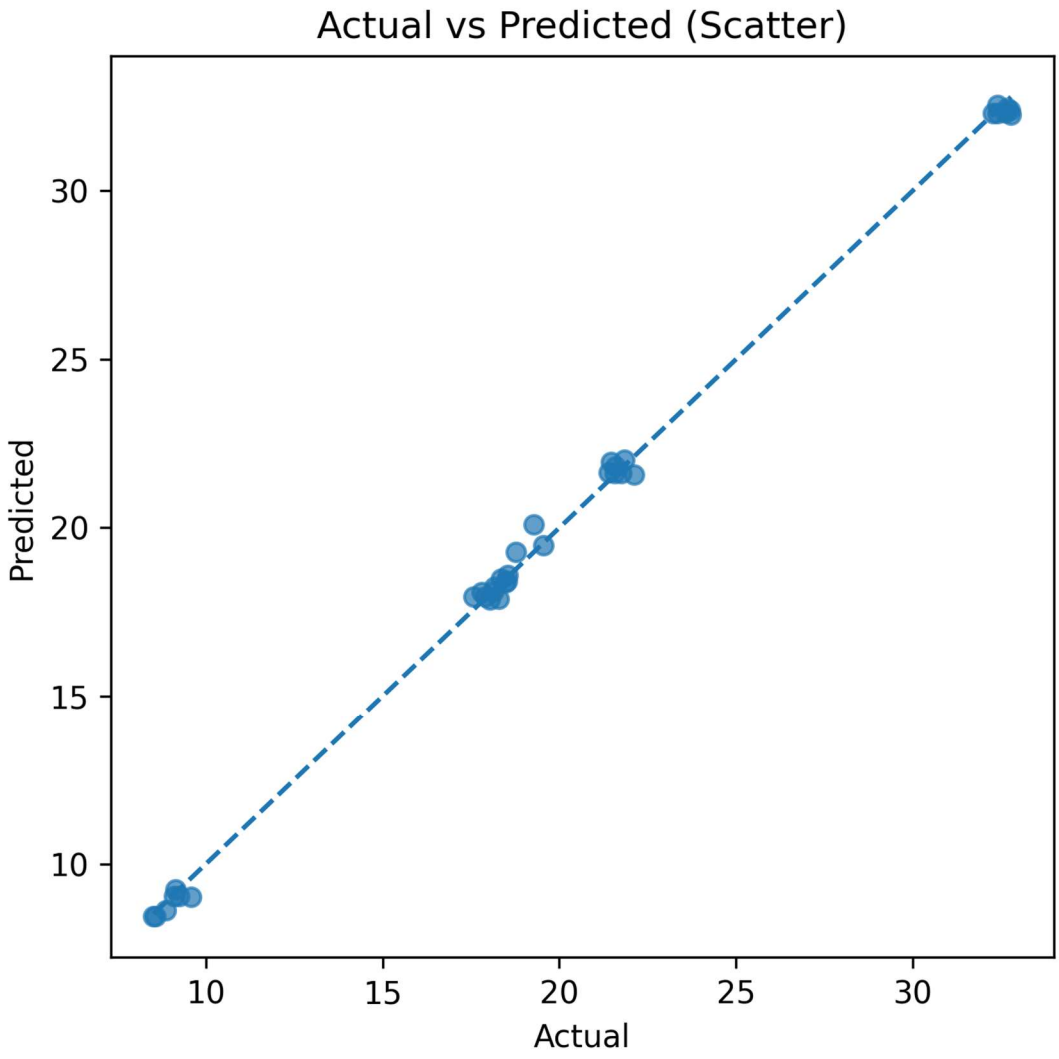

Figure P4-3: Actual vs predicted scatter

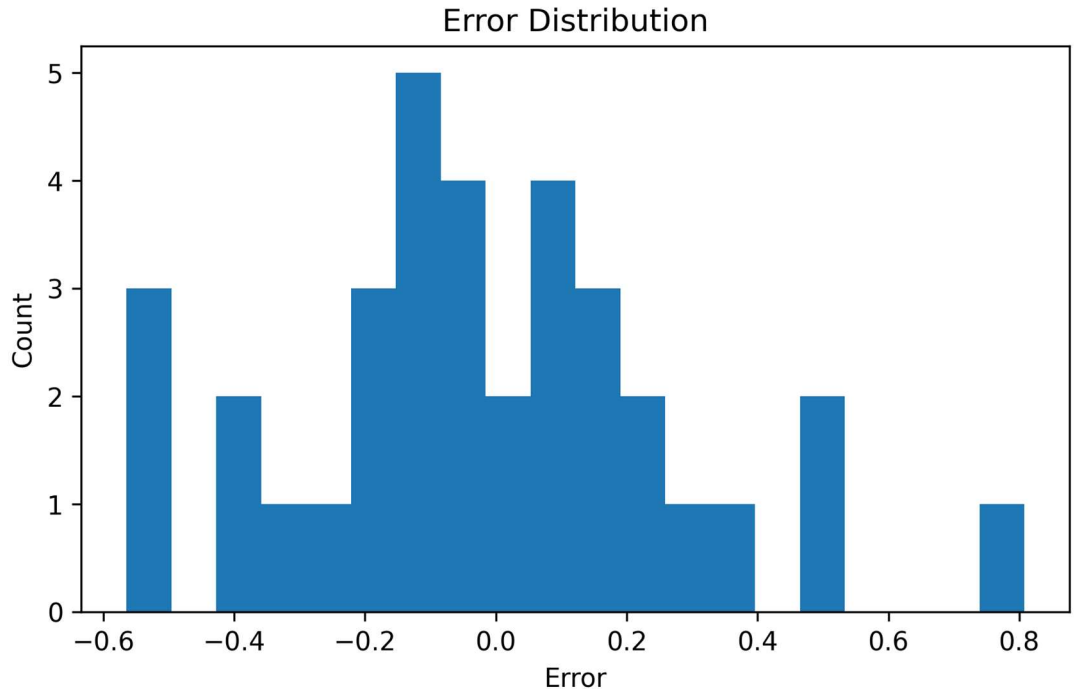

Figure P4-4: Error distribution

Business Value

Impact Quantification

Accurate market-share forecasting supports proactive pricing, marketing, and competitive responses.

Recommendations

- Refresh news embeddings monthly for drift control.
- Add competitor-level features for richer signal coverage.
- Deploy alerting for predicted share drops beyond threshold.

Limitations and Future Work

Current Limitations

- Model is trained on t+3 horizon only; multi-horizon expansion needed.
- NLP signals depend on tag coverage quality.

Future Improvements

- Extend to multi-horizon direct forecasting (t+1..t+12).
- Use transformer-based embeddings for richer news representation.

Technical Details

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Model Architecture

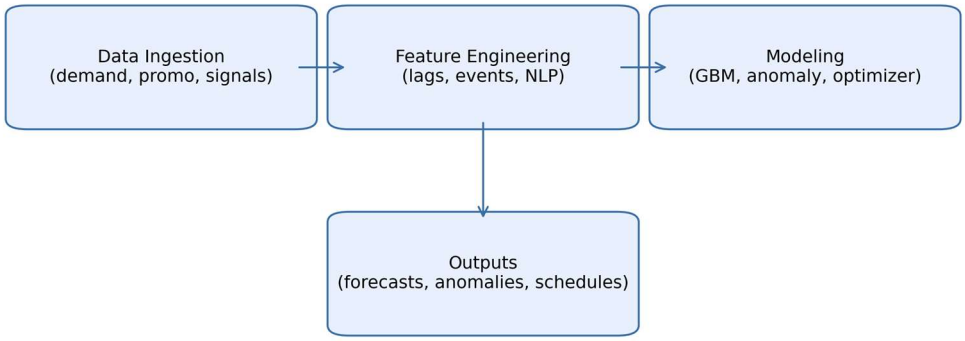

Figure: End-to-end architecture overview

Feature Importance

Visualization Gallery

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